

2025 Year 2 Semester 1

Plant Disease Detection using Computer Vision

IT2011- Artificial Intelligence and Machine Learning

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1. Introduction & Problem Statement

Plant diseases cause major losses in global agriculture, reducing crop yield and threatening food security. Traditional identification methods rely on expert inspection, which is time-consuming, costly, and often inaccurate in real farm environments. Early and accurate detection is essential to protect crops and support sustainable farming.

This project aims to develop and evaluate Machine Learning (ML) and Deep Learning (DL) models to automatically classify plant leaf diseases using image data from the PlantVillage dataset. The selected diseases include Potato Late Blight, Tomato Leaf Mold, Tomato Early Blight, and Pepper Bacterial Spot.

The objective is to compare the performance of multiple models MobileNetV2**,** DenseNet121, InceptionV3, VGG16, EfficientNetB0, KNN, and Logistic Regression for effective disease classification and to demonstrate how AI can enhance agricultural productivity through fast and reliable diagnosis.

2. Dataset Description

- Dataset Name**:** PlantVillage Dataset
- Source**:** Kaggle (Publicly available agricultural image dataset)
- Size: The dataset contains over 54,000 leaf images categorized across multiple plant species and disease types. For this project, a subset of 4 disease classes was selected for focused training and evaluation.
- Classes Used:
  - ✓ Potato Late Blight
  - ✓ Tomato Leaf Mold
  - ✓ Tomato Early Blight
  - ✓ Pepper Bacterial Spot
- Data Type**:** RGB color images of plant leaves (resized to 224×224 pixels).

• Target Variable**:** Disease label (multiclass categorical: 4 disease categories).

• Data Split:

- ✓ Training set: 80% of total images
- ✓ Validation set: 20% of total images

• Class Distribution**:** The dataset shows slight class imbalance, with Tomato Leaf Mold having the highest number of samples compared to the others.

• Key Features**:** Each image captures disease-specific leaf patterns such as:

- ▪ Texture*:* Spots, blights, and discoloration patterns
- ▪ Color Variations*:* Yellowing, browning, or mold regions
- ▪ Shape Patterns*:* Lesion shapes, holes, or edge damage

• Relevance**:** The dataset provides sufficient diversity in visual patterns for each disease class, making it ideal for training deep learning models (MobileNetV2, DenseNet121, VGG16, InceptionV3, EfficientNetB0) and classical models (KNN, Logistic Regression) for leaf disease classification.

• Limitations & Bias:

- ▪ Images are captured in controlled environments with plain backgrounds — different from real-field lighting.
- ▪ Slight imbalance among classes.
- ▪ Limited representation of environmental noise (e.g., shadows, overlapping leaves).

• Mitigation Techniques**:**

- ▪ Applied data augmentation (rotation, flipping, zooming, brightness adjustments) to increase dataset variability.
- ▪ Used balanced training batches to minimize class bias and improve generalization.

3. Preprocessing & EDA

Data Cleaning and Preparation:

• Invalid Image Removal**:** All image files were verified for accessibility, and any unreadable or corrupted images were removed to maintain dataset integrity.

• Image Resizing**:** Every image was resized to a uniform dimension of 224×224 pixels to match the input requirements of CNN-based models such as

MobileNetV2, DenseNet121, and EfficientNetB0.

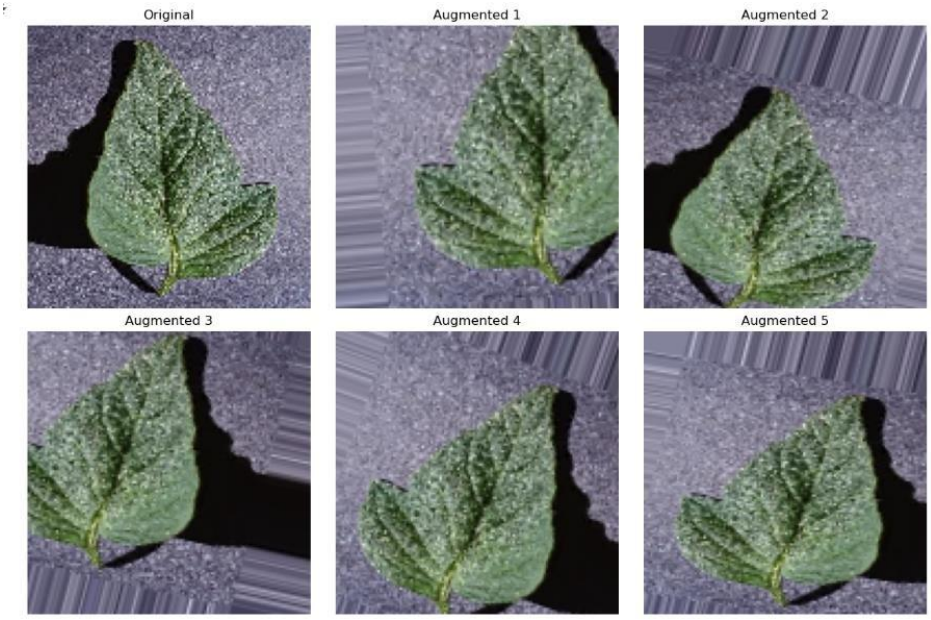
- Color Conversion**:** All images were converted to RGB format to ensure color consistency and compatibility with pre-trained models (some original images were in grayscale).

- Normalization**:** Pixel values were scaled from 0–255 to 0–1 using rescaling (rescale=1./255) to improve model convergence and numerical stability during training.

- Data Splitting**:** The dataset was divided into 80% training and 20% validation sets using Keras' ImageDataGenerator, enabling batch loading and automatic class labeling.

Data Augmentation:

To enhance the robustness of the model and minimize overfitting, data augmentation techniques were applied. The training generator included rotation (40°), zoom (0.3), width and height shifts (0.3), shear transformations, and horizontal and vertical flips. These transformations helped simulate real-world variations in leaf orientation and lighting, effectively balancing the dataset and increasing its diversity.



Exploratory Data Analysis (EDA):

• Class Distribution**:** A bar chart was created to visualize image counts per class, revealing a mild imbalance, with Tomato Leaf Mold having slightly more samples than other categories.

• Sample Visualization**:** A grid of sample images was plotted for each disease type (Potato Late Blight, Tomato Leaf Mold, Tomato Early Blight, and Pepper Bacterial Spot) to examine visual differences in color, texture, and lesion pattern.

4. Model Design & Implementation

The prediction task is a multiclass image classification problem aimed at detecting four plant leaf diseases (Potato Late Blight, Tomato Leaf Mold, Tomato Early Blight, and Pepper Bacterial Spot). Multiple deep learning and machine learning models were developed and evaluated on a standard 80/20 train-validation split.

Class Imbalance Handling

To address slight class imbalance in the dataset, data augmentation was applied to the training set using techniques such as rotation (40°), zoom (0.3), width/height shift (0.3), shear, and horizontal/vertical flips. This improved dataset diversity and prevented models from overfitting to majority classes.

Models Implemented

1. MobileNetV2 – Lightweight CNN using depthwise separable convolutions for efficient training and high accuracy on small datasets.
2. DenseNet121 – Deep CNN with dense connections that promote feature reuse and strong gradient flow.
3. VGG16 – Classic CNN baseline with sequential convolutional blocks for feature extraction.
4. InceptionV3 – Multi-scale architecture that captures diverse lesion features using parallel convolutional filters.
5. EfficientNetB0 – Modern, optimized CNN model balancing accuracy and computational cost through compound scaling.
6. K-Nearest Neighbors (KNN) – Traditional machine learning classifier used as a non-parametric baseline.

7. Logistic Regression – Linear model baseline applied on extracted deep features to test linear separability.

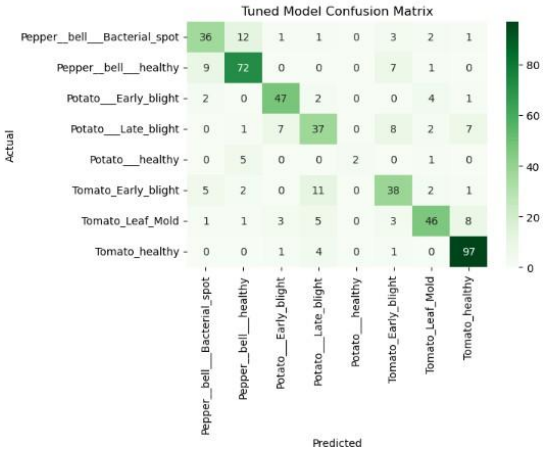
Training Details

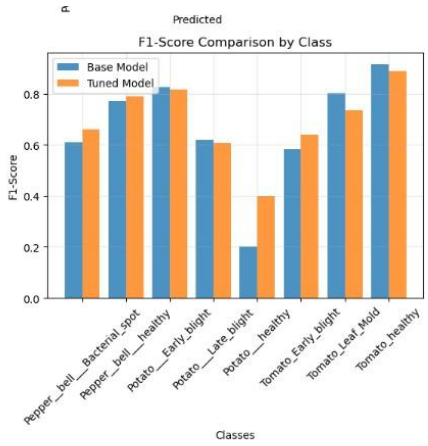
- Image Size**:** 224×224×3
- Normalization: Rescale pixel values to [0, 1]
- Batch Size: 32
- Optimizer / Loss: Adam / Categorical Crossentropy
- Epochs: 20–30 (with Early Stopping and ReduceLROnPlateau)
- Callbacks: EarlyStopping(patience=5), ReduceLROnPlateau(patience=3)
- Hardware: GPU runtime (TensorFlow/Keras environment)

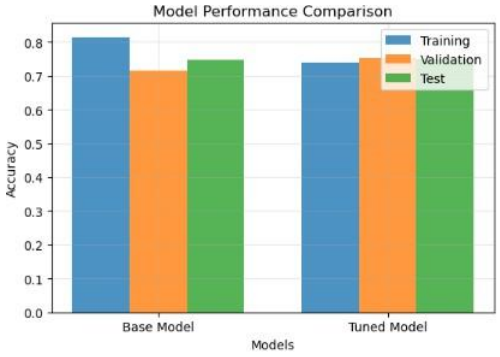
5. Evaluation and Comparison

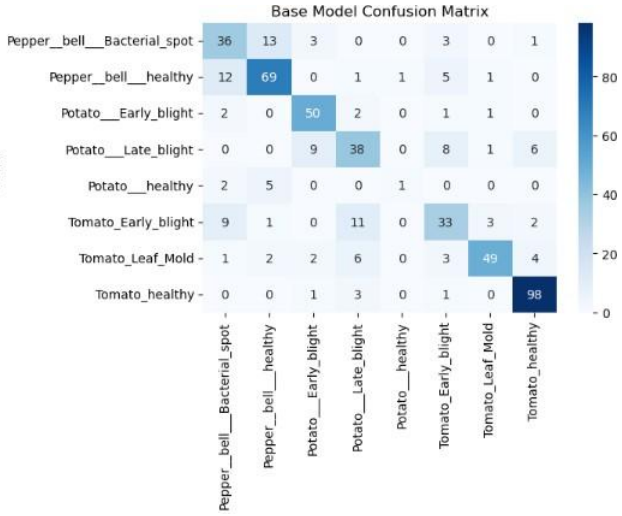
Model	**Precision**	**Recall**	**F1-Score**	**Accuracy**
InceptionV3	0.9878	0.9878	0.99	0.9878
DenseNet121	0.9855	0.9853	0.99	0.9853
MobileNetV2	0.9806	0.9804	0.98	0.9804
VGG16	0.9405	0.9351	0.94	0.9351
KNN	0.7711	0.7686	0.77	0.7686
Logistic Regression	0.6258	0.6212	0.62	0.6212
EfficientNetB0	0.19	_	_	_

These figures llustrate evaluation results obtained from the KNN model highlighting the improvements achieved through hyperparameter tuning and performance analysis.









Evaluation Summary:

All models were evaluated using Precision, Recall, F1-Score, and Accuracy to measure their predictive performance on the validation dataset. These metrics provide a balanced understanding of how well each algorithm classified plant leaf diseases.

presents the summarized results. Among the tested models, InceptionV3 and DenseNet121 achieved the highest overall performance, followed closely by MobileNetV2 (0.9804). These models demonstrated strong generalization and feature extraction capabilities, confirming the effectiveness of transfer learning in visual recognition tasks.

VGG16 also performed well (Accuracy = 0.9351), though with slightly reduced precision due to its higher parameter count. Traditional models such as KNN (0.7686) and Logistic Regression (0.6212) showed noticeably lower performance, indicating the limitations of shallow learners in handling complex image features. EfficientNetB0 showed unstable results due to incomplete tuning and data configuration errors during training.

6. Ethical Considerations & Bias Mitigation

Ethical Considerations:

- Fairness & Responsibility: Ensured AI models make unbiased predictions across all disease classes and do not favor majority samples
- Dataset Bias: The PlantVillage dataset was collected in controlled lab environments with clean backgrounds, which may not represent real farm conditions.

- Transparency: Model predictions must be interpretable and explainable to avoid misuse or blind reliance by users.
- Ethical Data Usage: The dataset was used only for academic research and respects open-source licensing guidelines.

Bias Identification:

- Data Augmentation**:** Applied rotation, zoom, flipping, and brightness adjustments to balance dataset diversity.
- Class Weighting**:** Used class weights during model training to reduce bias toward dominant classes.
- Fair Evaluation: Metrics such as Precision, Recall, and F1-Score were used instead of only Accuracy to ensure fair performance across all classes.
- Explainability: Implemented Grad-CAM visualizations to interpret which leaf regions influenced model predictions.

7. Reflection & Lessons Learned

This project enhanced both technical and teamwork skills. Each member handled different models and tasks, improving coordination and problem-solving.

We gained practical experience in transfer learning, preprocessing, and evaluation metrics, and learned how data augmentation and class weighting reduce bias and improve fairness.

Working with deep learning frameworks also strengthened our understanding of model training and optimization.

Challenges included limited GPU resources and handling large image datasets efficiently. These were overcome through early stopping, reduced batch sizes, and collaboration.

Future Improvements:

- Improve Model Efficiency**:** Apply model compression or pruning to reduce overfitting and make CNNs more suitable for low-power devices.

- Optimize Training Time**:** Use GPU-based cloud platforms to overcome limited hardware issues and speed up model training.

8. References

- Kaggle. (2020). PlantVillage Dataset. Retrieved from https://www.kaggle.com/datasets/emmarex/plantdisease

- TensorFlow Developers. (2023). TensorFlow and Keras Documentation. Retrieved from https://www.tensorflow.org/